Optimisation Theory & Reliability

Oral Re-examination August 2022 - Procedure and Questions

Rules *- please read these carefully!*

1. The total oral examination time is 20 minutes per student, ***including*** *time for voting. The actual examination time is therefore approximately 15. The students should therefore prepare so he/she is able to answer the drawn questions well within this time frame.*

2. Each student must answer questions from both the optimisation and reliability part of the course. The two questions will be drawn from two pools of questions, pool 1: optimisation theory (questions 1-6) and pool 2: reliability(questions 7-13).

3. There is no time allocated for preparation; the examination starts immediately after the questions have been drawn by the student.

4. The questions included in this document are the foundation for the examination. The examiners may/will however ask small supplementary questions.

5. Notice that emphasis is on understanding the concepts/theory, not necessarily on completing all calculations.

6. The students are allowed to bring papers to the examination. It is however expected that the students will be able to answer the questions, without looking in the papers!

7. All questions are included in this document, which is forwarded to the students at least one week before the day of the reexamination.

Question 1: Constrained minimisation problem with 2 variables

The following constrained minimisation problem is considered:

minimize $f(\mathbf{x}) \;=\; (x_1 + 1)^2 + (1 - x_2)^2$

Subject to the inequality constraints:

$g_1(\mathbf{x}) \;=\; x_1 + 0.5x_2 - 2 \leq 0$

$g_2(\mathbf{x}) \;=\; -x_1 + 1 \leq 0$

Determine the optimum solution to the problem using the KKT necessary conditions and/or graphical methods.

Question 2: Steepest descent method

We will consider gradient-based minimisation of the following unconstrained function:

$f(\mathbf{x}) \ = \ (-x_1 + x_2)^2 + 2x_1 + 1$

The starting point is: $\mathbf{x}^{(0)} = [1,\ -1]^T$.

a) In order to complete the first iteration of the steepest descent method for the function using the analytical approach, the gradient must be evaluated. What is the expression for the gradient, $\mathbf{c}(\mathbf{x})$?

b) In order to complete the first iteration of the steepest descent method for the function above using the analytical approach, the 1D line search problem should be established. What is the expression for $f(\alpha)$, where α is the step size in direction $\mathbf{d}^{(0)}$?

c) Complete the first iteration of the steepest descent method for the function above. What are the values of the design variables after the first iteration, $\mathbf{x}^{(1)}$?

Question 3: Newton's method

We will consider gradient-based minimisation of the following unconstrained function:

$$f(\mathbf{x}) \;=\; (x_1 - x_2)^2 + (x_1 - 2)^2 + 7$$

The starting point is: $\mathbf{x}^{(0)} = [-1, 1]^T$.

a) In order to complete the first iteration of the modified Newton's method for the function, the Hessian must be evaluated. What is the expression for the Hessian, $\mathbf{H}(\mathbf{x})$?

b) In order to complete the first iteration of the modified Newton's method for the function above using the analytical approach, the 1D line search problem should be established. What is the expression for $f(\alpha)$, where α is the step size in direction $\mathbf{d}^{(0)}$?

c) Complete the first iteration of the steepest descent method for the function above. What are the values of the design variables after the first iteration, $\mathbf{x}^{(1)}$?

Question 4: Linear Simplex Method

A linear optimisation problem is given by:

$$\text{minimise} \qquad f(\mathbf{x}) = -2x_1 - 10x_2$$

Subject to the constraints:

$$\begin{array}{rcccc} g_1(\mathbf{x}) & = & -x_1 + x_2 & \leq & 4 \\ g_2(\mathbf{x}) & = & x_1 + x_2 & \leq & 6 \\ g_3(\mathbf{x}) & = & -2x_1 - x_2 & \geq & -10 \\ & & x_i \geq 0 & \forall & x_i = \{1, 2\} \end{array}$$

a) Describe the system using the standard linear problem definition.

b) Determine the extended tableau for the problem.

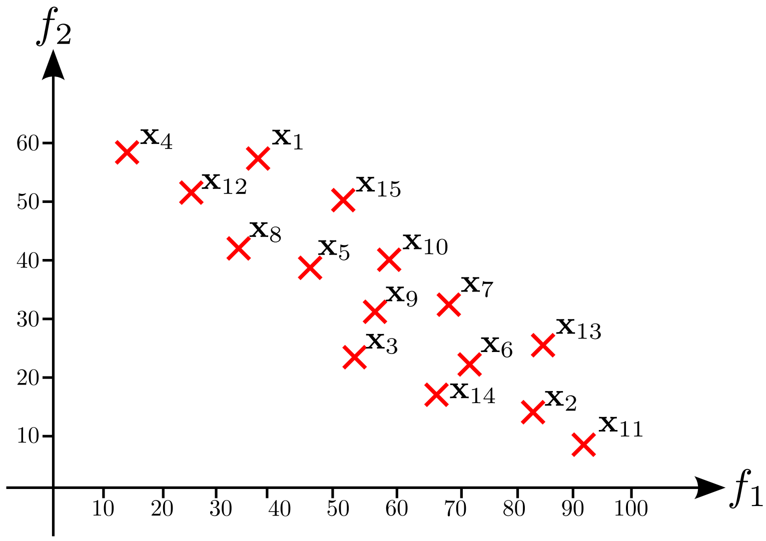
c) Determine the pivot column and row for the first iteration.

d) Solve the optimization problem and determine which of the variables that are basic variables.

e) Determine the effect of changing each of the different resource limits by 1.

Question 5: Pareto Optimality and Multi-Objective Optimization

The below figure shows a number of points from an multi-objective optimisation. It is informed that the non-dominated points are part of the Pareto front.



a) Determine the points that are Pareto optimal and explain why this is the case, i.e. what defines Pareto optimality?

b) Determine the Utopia point, when assuming that the above Pareto points are a good representation for the extremes of the Pareto front.

c) Determine whether each of the followings statements are correct or not. Explain why this is so:

1. The simulated annealing method does not require the objective function to be continuous
2. For multi-objective problems it is always possible to determine the Utopia point
3. For multi-objective problems the Utopia point is the best possible design one can obtain if the optimisation is running for a sufficient number of iterations
4. Genetic algorithms are derived from ideas from evolution theory and do not require the calculation of gradients of the objective function
5. Inaccurate line search methods may be used as an alternative to golden section search in all line search based methods
6. A multi-objective optimization problem will always have one global minimum

Question 6: Alternate KKT conditions

An optimisation problem is formulated as:

$$\text{minimise} \qquad f(\mathbf{x}) = (x_1 - 2)^2 + (x_2 - 4)^2 - 2x_1x_2$$

Subject to:

$$\begin{aligned} g_1(\mathbf{x}) &= 5x_1 - 4x_2 + 8 \leq 0 \\ g_2(\mathbf{x}) &= 2x_1 + x_2 + 14 \leq 0 \end{aligned}$$

a) Using the alternate form of the KKT necessary conditions determine the gradient conditions, switching conditions, feasibility check and non-negativity of the Lagrange multipliers.

b) Determine if there are any candidate points lying on the constraint $g_2(\mathbf{x})$, which fulfills the necessary KKT conditions, and if so determine the values of the Lagrange multipliers and constraints at this/these points.

c) Determine whether each of the followings statements are correct or not. Explain why this is so:

1. It is always possible to solve a non-linear problem using a SLP method
2. A SQP approach will always yield faster convergence than a SLP method
3. If converging a SQP approach will always yield the global minimum
4. It is always possible to find the optimum for a linear optimisation problem
5. SLP methods do not rely on a descent function, why continuous descent cannot be guaranteed
6. Quasi-Newton methods require that you can determine the Hessian analytically

Question No. 7

A lever of a car clutch is guaranteed for 4,000,000 operational cycles with normal $\Delta T = 80$ °C. The number of cycle to failure versus ΔT follows an inverse power law with a power of -4.10 .

Find the equivalent end of life at severe operation condition of $\Delta T = 90$ °C, assuming 100 cycles per day.

Question No. 8

An electronic power supply for laptop PC comprises:

A. a main fuse
B. a 4-diode main rectifier
C. 3 dc-link capacitors in parallel
D. 2 main power switches
E. a transformer
F. a 2-diode secondary rectifier
G. an output filter
H. a power indicator LED

Assumptions:

1. All the above parts but the LED are needed to operate correctly.
2. Diodes 1 and 3 or diodes 2 and 4 are sufficient to make the main rectifier operate correctly.
3. 2 dc-link capacitors out of 3 are sufficient to keep the power supply working;
4. Both main switches are needed to properly operate;
5. The transformer and the output filter can be supposed to have infinite lifetime;
6. One diode in the secondary rectifier is sufficient to properly operate;
7. all the remaining parts (e.g. power cord, plastic enclosure, laptop cord & plug) are supposed to have infinite lifetime in respect to the ones listed above.

Sketch up the reliability block diagram of the system.

Question No. 9

Which system of the ones with reliability functions listed below would fail with higher probability after 1-year operation?

A – Exponential, $\lambda = 20.0$ failures per million hours

B – Weibull, $\beta = 1.5$, $\eta = 30.000$ hours

Question No. 10

The bolts of a horizontal heat sink experience cyclic stress at the base-plate connection due to the thermal expansion of the heat sink itself.

Accelerated data was taken on bolts where a cyclical stress was applied to the bolts whereby the metal at the base plate came under a continuous cyclical stress range of -4.0 MPa to +4.0 MPa with a mean tensile stress of zero. The bolts failed at the base plate connection after 10,000 cycles. Assuming a power-law exponent of m = 3.5 for the cycling, a metal tensile strength of 7.4 MPa and no defined elastic range, estimate the number of cycles-to-failure CTF that would be expected in real use if the stress is between -2.5 MPa to +2.5 MPa.

Question No. 11

A shaft was tested in the laboratory to determine its fatigue life. The test results were as follows:

Stress level ($\times 10^2$ Nm)	3.4	7.2	12.0
Mean cycles to failure ($\times 10^6$)	12.0	5.4	0.85

The shaft will operate in a real motor with the most relevant stress levels occurring in the following proportions, respectively:

Proportion of cycles	0.6	0.3	0.1

a) In case of 100 cycles per hour, what will be the expected time to failure in service?

Question No. 12

Ten commercial LED lamps failed at different times: 3750d, 2500d, 1450d, 878d, 4012d, 2704d, 1830d, 3121d, 5608d and 1701d.

1) State the relationship between accumulated failure and time in Weibull distribution.
2) Arrange these time-to-failure numbers by using median ranking method.
3) Plot the ranking number as listed in (b) and the cycles to failure in the attached Weibull paper below. Please find out the values for β and η, respectively.
4) Explain how to identify the values of β and η in the Weibull plot.

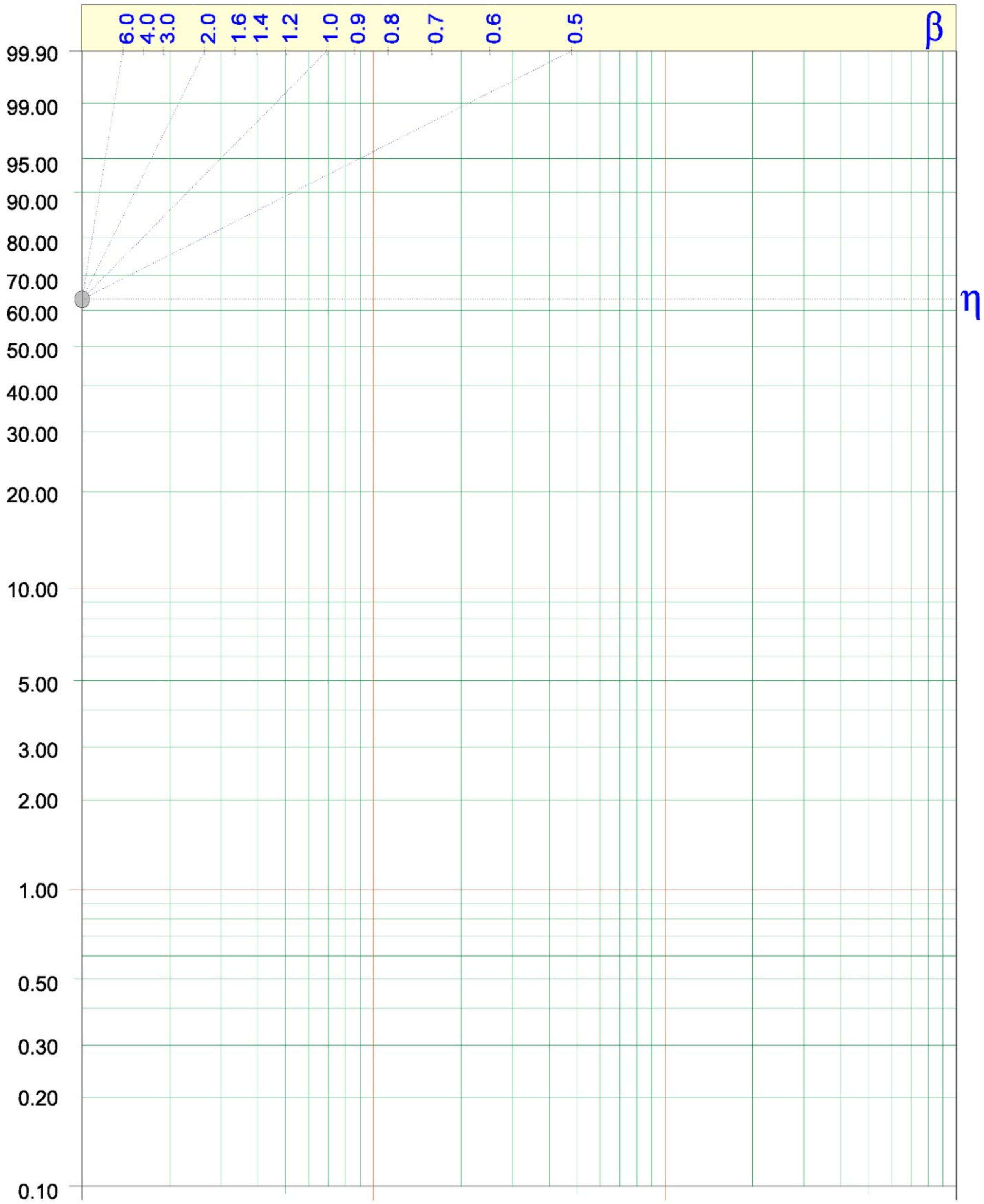
Appendix I – Median rank table

sample size = n

failure rank = i

n

i	1	2	3	4	5	6	7	8	9	10
1	.5000	.2929	.2063	.1591	.1294	.1091	.0943	.0830	.0741	.0670
2		.7071	.5000	.3864	.3147	.2655	.2295	.2021	.1806	.1632
3			.7937	.6136	.5000	.4218	.3648	.3213	.2871	.2594
4				.8409	.6853	.5782	.5000	.4404	.3935	.3557
5					.8706	.7345	.6352	.5596	.5000	.4519
6						.8906	.7705	.6787	.6065	.5481
7							.9057	.7979	.7129	.6443
8								.9170	.8194	.7406
9									.9259	.8368
10										.9330



6.0
4.0
3.0
2.0
1.6
1.4
1.2
1.0
0.9
0.8
0.7
0.6
0.5
β
η
99.90
99.00
95.00
90.00
80.00
70.00
60.00
50.00
40.00
30.00
20.00
10.00
5.00
3.00
2.00
1.00
0.50
0.30
0.20
0.10

Question No. 13

Plot a rainflow diagram of a bolt operating under the following stress sequence:
0,50,20,40,25,35,22,50,28,48,42,40,35,40,25,60,40,20,30 [MPa]